Cisco Customer Experience

Norwegian Cruise Line
Method of Procedure

Fusion Routing Changes for Macro-segmentation

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Version 1.0

Cisco Systems, Inc.
Corporate Headquarters

170 West Tasman Drive
San Jose, CA 95134-1706 USA

Phone: +1 408-526-4000
Toll Free: +1 800-553-NETS (6387)
Fax: +1 408-526-4100

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About This Document

Rendered by	Javier Sánchez (javiersa@cisco.com), Omar Candarave (ocandara@cisco.com), Xiaoning Zhang (harrzhan@cisco.com)
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Document Conventions

Tip
Time saver. Expedite the task by following the recommendation being described.

Info
Alerts reader that the information will help them solve a problem or better understand the subject being described.

Note
Alerts readers to be careful. You might do something that could negatively impact a solution, project, equipment, or the quality of the work being described.

Warning
Alerts readers of a situation that could cause injury or severely impact a solution, project, equipment, or the quality of the work being described.

1 Introduction

1.1 Preface

Norwegian Cruise Line has implemented an SD-Access Fabric for their Campus Network and ACI For the Data Center in their newest cruise ship “PRIMA”. To enforce SD-Access fabric macro-segmentation and deny inter-VN communication, some routing changes are required to be made to the current Fusion Router configurations. These changes also provide a framework to be used in the future when NCL decides to permit inter-VN route leaking but still enforce segmentation as needed using SGTs.

The purpose of this document is to provide the technology-specific configuration and implementation instructions necessary to carry out the Fusion Router changes.

1.2 Audience

This document is intended for use by Norwegian Cruise Line and Cisco, including:

- Norwegian Cruise Line IT team
- Cisco CX project team

1.3 Scope

The scope of this document is described below.

In scope:

- Fusion Router changes to extend macro-segmentation from the SD-Access fabric.

1.4 Requirements

The following requirements have been discussed with Norwegian Cruise Line stakeholders during the design workshops for this use case:

- Devices in one Virtual Network (VN) **must not** be able to communicate to devices in a different VN, by default.
- In case needed, inter-VN communication can be allowed through route leaking.

The following image details proposed design, segmenting every VN/VRF and exporting/importing the required routes to/from the GRT to then reach the ACI fabric:

Proposed Design

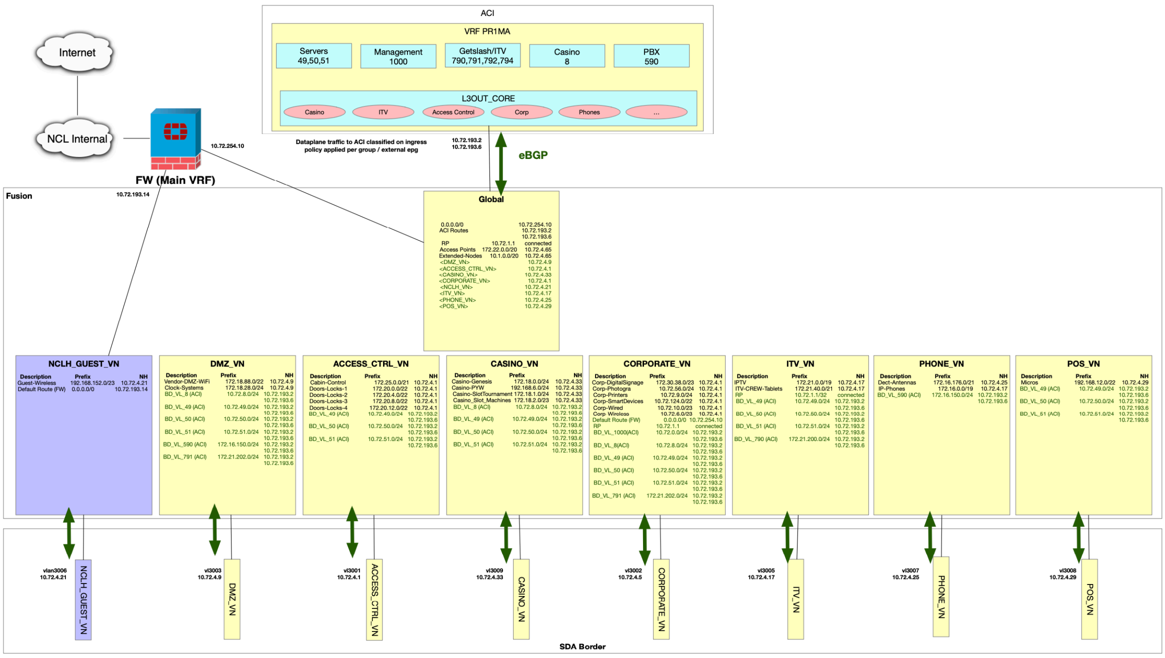

Figure 1 Macro-segmentation Proposed Design

1.5 Related Documents

The following documents have been used as an input or reference for this document.

Table 1 Related Documents

No.	Document Title	Document URL (If Any)
1	950233-NCL-DNAC-PDQ_1.0-ANSWERED	https://ciscoshare.cisco.com/alfext/ui/c/file/preview/70a74f96-1ce1-4c72-a2b2-bd16a07c883f
2	Norwegian Cruise Line - SDA - SRD-v1.0-20210612_101133	https://dcpcs.cloudapps.cisco.com/dcpcs/download.jsp?contentId=1353821&language=en-us
3	Norwegian Cruise Line - SDA - Solution Design Document-v1.0-20210723_094048	https://dcpcs.cloudapps.cisco.com/dcpcs/download.jsp?contentId=1435753&language=en-us
4	Norwegian Cruise Line - SDA - Test Plan-v1.0-20211005_145853	https://dcpcs.cloudapps.cisco.com/dcpcs/download.jsp?contentId=1471509&language=en-us
5	Norwegian Cruise Line - SDA - NIP-v1.0	https://dcpcs.cloudapps.cisco.com/dcpcs/download.jsp?contentId=1487781&language=en-us
6	A51089_NCL_SDA_INAUGURALS UC No1_MOP_v1.0_20220817	https://dcpcs.cloudapps.cisco.com/dcpcs/download.jsp?contentId=1602660&language=en-us

2 Fusion Router Changes

The following are the steps to follow. Most of the configurations are performed in the Fusion Router. Configurations performed on the other devices are noted accordingly.

The images added are just for reference purposes, generated from a Cisco Lab. Even though efforts were made to resemble NCL's PRIMA environment as much as possible, there might be slight differences (IPs, hostnames, names in the Hierarchy, number of Fabric Zones, among others).

2.1 Preparation

The following tasks should be performed prior to carry out the changes:

- Secure a change window. The changes documented in this MOP require a temporary network-down outage for devices below the Fusion layer in SD-Access. All the endpoints and devices will lose upstream connectivity for a short time. NCL is recommended to notify users and application owners accordingly.
- Because the expected link disruption between Fusion and Border, it is mandatory to secure console connections to Fusion and Border that do not depend on Fusion and Border connectivity.
- Backup Cisco DNA Center.
- Verify that Fusion Router and Border Router are healthy, and both are in Managed state.
- Perform a "sh ip route vrf" on each VRF and save the output to a file to a local computer so that they can easily be used for comparison, in case needed.
- Save the current running config to the startup file.
- Backup the configurations of Fusion Router and Border. Save these backup files to a local computer so that they can easily be used for comparison, in case needed.
- Execute the following command to save a copy of the current running-configuration in the bootflash of the Fusion Router:

```
copy running-config bootflash:Pre-Macro-segmentation-changes.txt
```

2.2 Create VRFs

From a console connection to the Fusion Router, create the VRFs to match the existing VNs in the SD-Access Border:

```
vrf definition ACCESS_CTRL_VN
rd 1:4100
!
address-family ipv4
route-target export 1:4100
route-target import 1:4100
exit-address-family
!
vrf definition CORP_VN
```

```
rd 1:4105
!
address-family ipv4
route-target export 1:4105
route-target import 1:4105
exit-address-family
!
vrf definition IPTV_VN
rd 1:4099
!
address-family ipv4
route-target export 1:4099
route-target import 1:4099
exit-address-family
!
vrf definition PHONE_VN
rd 1:4102
!
address-family ipv4
route-target export 1:4102
route-target import 1:4102
exit-address-family
!
vrf definition POS_VN
rd 1:4101
!
address-family ipv4
route-target export 1:4101
route-target import 1:4101
exit-address-family
!
vrf definition CASINO_VN
rd 1:4107
!
address-family ipv4
route-target export 1:4107
route-target import 1:4107
exit-address-family
!
```

Section 2.3 – 2.6 reflect a route import/export approach requested by NCL such that all global routes are leaked into each VRF and only leverages ACI contracts to control the communication in between these routes. This approach simplifies configurations in Fusion and does not require Fusion modifications in the event of subnet changes. The planned configuration changes will follow this method. It should be noted that this approach loses the granular route control capabilities inside Fusion. In the future when NCL does require inter-VRF route leaking, route maps need to be re-configured. For NCL's reference, the Cisco suggested route import/export strategy is documented in Appendix A. An example of inter-VRF route leaking is documented in Appendix B.

2.3 Create IP prefix-lists to import from GRT to each VN

Enter the following to the Fusion Router:

```
!
ip prefix-list PERMIT-ALL-BUT-Q0 seq 5 deny 0.0.0.0/0
ip prefix-list PERMIT-ALL-BUT-Q0 seq 10 permit 0.0.0.0/0 le 32
!
ip prefix-list PERMIT-ALL seq 5 permit 0.0.0.0/0 le 32
!
```

2.4 Create Route-maps to export to GRT

Enter the following to the Fusion Router:

```
!
route-map PHONE-TO-GRT permit 10
match ip address prefix-list PERMIT-ALL
!
route-map ACCESS_CTRL-TO-GRT permit 10
match ip address prefix-list PERMIT-ALL
!
route-map IPTV-TO-GRT permit 10
match ip address prefix-list PERMIT-ALL
!
route-map CASINO-TO-GRT permit 10
match ip address prefix-list PERMIT-ALL
!
route-map CORP-TO-GRT permit 10
match ip address prefix-list PERMIT-ALL
!
route-map POS-TO-GRT permit 10
match ip address prefix-list PERMIT-ALL
!
```

2.5 Create Route-maps to import prefix-lists from GRT

Enter the following to the Fusion Router:

```
!
route-map IMPORT-TO-CORP permit 10
match ip address prefix-list PERMIT-AL
!
route-map IMPORT-TO-IPTV permit 10
match ip address prefix-list PERMIT-ALL-BUT-Q0
!
route-map IMPORT-TO-PHONE permit 10
match ip address prefix-list PERMIT-ALL-BUT-Q0
!
route-map IMPORT-TO-CASINO permit 10
match ip address prefix-list PERMIT-ALL-BUT-Q0
!
route-map IMPORT-TO-POS permit 10
match ip address prefix-list PERMIT-ALL-BUT-Q0
!
route-map IMPORT-TO-ACCESS_CTRL permit 10
match ip address prefix-list PERMIT-ALL-BUT-Q0
```

2.6 Associate required route-maps to each VRF

Enter the following to the Fusion Router:

```
vrf definition ACCESS_CTRL_VN
!
address-family ipv4
import ipv4 unicast map IMPORT-TO-ACCESS_CTRL
export ipv4 unicast map ACCESS_CTRL-TO-GRT
exit-address-family
!
vrf definition CORP_VN
!
address-family ipv4
import ipv4 unicast map IMPORT-TO-CORP
export ipv4 unicast map CORP-TO-GRT
exit-address-family
!
vrf definition IPTV_VN
!
address-family ipv4
import ipv4 unicast map IMPORT-TO-IPTV
export ipv4 unicast map IPTV-TO-GRT
exit-address-family
!
vrf definition PHONE_VN
!
address-family ipv4
import ipv4 unicast map IMPORT-TO-PHONE
export ipv4 unicast map PHONE-TO-GRT
exit-address-family
!
vrf definition POS_VN
!
address-family ipv4
import ipv4 unicast map IMPORT-TO-POS
export ipv4 unicast map POS-TO-GRT
exit-address-family
!
vrf definition CASINO_VN
!
address-family ipv4
import ipv4 unicast map IMPORT-TO-CASINO
export ipv4 unicast map CASINO-TO-GRT
exit-address-family
!
```

2.7 BGP peering changes

Enter the following to the Fusion Router:

```
router bgp 65000
!
address-family ipv4
maximum-paths 2
exit-address-family
!
address-family ipv4 vrf ACCESS_CTRL_VN
neighbor 10.72.4.1 remote-as 65001
```

```
neighbor 10.72.4.1 update-source Vlan3001
maximum-paths 2
exit-address-family
!
address-family ipv4 vrf CASINO_VN
bgp aggregate-timer 0
neighbor 10.72.4.33 remote-as 65001
neighbor 10.72.4.33 update-source Vlan3009
neighbor 10.72.4.33 activate
maximum-paths 2
exit-address-family
!
address-family ipv4 vrf CORP_VN
bgp aggregate-timer 0
neighbor 10.72.4.5 remote-as 65001
neighbor 10.72.4.5 update-source Vlan3002
maximum-paths 2
exit-address-family
!
address-family ipv4 vrf IPTV_VN
bgp aggregate-timer 0
neighbor 10.72.4.17 remote-as 65001
neighbor 10.72.4.17 update-source Vlan3005
neighbor 10.72.4.17 activate
maximum-paths 2
exit-address-family
!
address-family ipv4 vrf PHONE_VN
bgp aggregate-timer 0
neighbor 10.72.4.25 remote-as 65001
neighbor 10.72.4.25 update-source Vlan3007
neighbor 10.72.4.25 activate
maximum-paths 2
exit-address-family
!
address-family ipv4 vrf POS_VN
bgp aggregate-timer 0
neighbor 10.72.4.29 remote-as 65001
neighbor 10.72.4.29 update-source Vlan3008
neighbor 10.72.4.29 activate
maximum-paths 2
exit-address-family
!
```

2.8 Associate SVIs to VRFs accordingly

Enter the following to the Fusion Router:

```
interface Vlan3001
vrf forwarding ACCESS_CTRL_VN
ip address 10.72.4.2 255.255.255.252
!
interface Vlan3002
vrf forwarding CORP_VN
ip address 10.72.4.6 255.255.255.252
ip pim sparse-mode
!
interface Vlan3005
vrf forwarding IPTV_VN
ip address 10.72.4.18 255.255.255.252
ip pim sparse-mode
!
```

```
interface Vlan3007
vrf forwarding PHONE_VN
ip address 10.72.4.26 255.255.255.252
!
interface Vlan3008
vrf forwarding POS_VN
ip address 10.72.4.30 255.255.255.252
!
interface Vlan3009
vrf forwarding CASINO_VN
ip address 10.72.4.34 255.255.255.252
!
```

It was identified that in PRIMA the SVI 210 is the one used to peer through the GRT between the Fusion Router and SD-Access Border. This SVI is meant to be used temporarily for initial discovery of devices only. The permanent SVI for this peering should be Vlan3004 (INFRA_VN) configured above. Once configured, execute the following:

1. Review BGP neighborship in Fusion Router:

```
[Current status, before adding an IP back to SVI 3004]

---output ommitted---
PRIMAFUSION#    sh ip bgp summ
BGP router identifier 10.72.1.1, local AS number 65000
Neighbor        V    AS   MsgRcvd MsgSent TblVer InQ OutQ Up/Down State/PfxRcd
10.72.4.13      4  65001     0       0        1    0    0 never    Idle
10.72.4.65      4  65001  74871   74891     360    0    0 6w5d            42
```

2. Add IP back to SVI 3004:

```
!
interface Vlan3004
description INFRA-VN
ip address 10.72.4.14 255.255.255.252
no ip redirects
ip route-cache same-interface
!
```

3. Review BGP peering for INFRA_VN:

```
[Expected output, once the IP Address is added back to SVI 3004]

---output ommitted---
PRIMAFUSION#    sh ip bgp summ
BGP router identifier 10.72.1.1, local AS number 65000
Neighbor        V    AS   MsgRcvd  MsgSent TblVer  InQ OutQ Up/Down State/PfxRcd
10.72.4.13      4  65001  74861    74861   360     0    0    0:05   42
10.72.4.65      4  65001  74871    74891   360     0    0    6w5d   42
```

4. Delete SVI 210 and BGP peering from Fusion and Border:

From Fusion:

```
!
no int vlan 210
no vlan 210
!
router bgp 65000
```

```
 no  neighbor 10.72.4.65 remote-as 65001
!
```

From Border:

```
!
no int vlan 210
no vlan 210
!
router bgp 65001
 no  neighbor 10.72.4.66 remote-as 65000
!
```

2.9 Remove current BGP peering from GRT

Enter the following to the Fusion Router:

```
router bgp 65000
no neighbor 10.72.4.1 remote-as 65001
no neighbor 10.72.4.5 remote-as 65001
no neighbor 10.72.4.17 remote-as 65001
no neighbor 10.72.4.25 remote-as 65001
no neighbor 10.72.4.29 remote-as 65001
no neighbor 10.72.4.33 remote-as 65001
```

2.10 Multicast migration to extranet

In Fusion, add the following configuration to configure multicast extranet for the required VRFs (IPTV_VN and CORP_VN):

```
ip multicast-routing vrf CORP_VN
ip multicast-routing vrf IPTV_VN

ip pim vrf IPTV_VN rp-address 10.72.1.1
ip pim vrf CORP_VN rp-address 10.72.1.1
```

2.11 Validation

From Fusion Router:

1. Verify all BGP neighborships are up for GRT:

```
C9500-Fusion#show ip bgp summary
BGP router identifier 10.72.192.1, local AS number 65000
BGP table version is 306, main routing table version 306
50 network entries using 12400 bytes of memory
60 path entries using 8160 bytes of memory
10 multipath network entries and 20 multipath paths
17/17 BGP path/bestpath attribute entries using 5032 bytes of memory
2 BGP AS-PATH entries using 48 bytes of memory
9 BGP extended community entries using 216 bytes of memory
0 BGP route-map cache entries using 0 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
BGP using 25856 total bytes of memory
```

```
BGP activity 3023/2806 prefixes, 4932/4645 paths, scan interval 60 secs
54 networks peaked at 18:19:20 Sep 6 2022 UTC (1w1d ago)

Neighbor        V           AS MsgRcvd MsgSent   TblVer  InQ OutQ Up/Down  
State/PfxRcd
10.72.4.13      4        65001    3236    3298      306    0    0 2d00h           3
10.72.193.2     4        65072    2948    3298      306    0    0 2d00h          10
10.72.193.6     4        65072    2948    3293      306    0    0 2d00h          10
```

2. Verify all BGP neighborships are up for each VRF.

ACCESS_CTRL_VN:

```
C9500-Fusion#show ip bgp vpnv4 vrf ACCESS_CTRL_VN summary
BGP router identifier 10.72.192.1, local AS number 65000
BGP table version is 1969, main routing table version 1969
25 network entries using 6400 bytes of memory
35 path entries using 4760 bytes of memory
10 multipath network entries and 20 multipath paths
25/24 BGP path/bestpath attribute entries using 7800 bytes of memory
2 BGP AS-PATH entries using 48 bytes of memory
9 BGP extended community entries using 216 bytes of memory
0 BGP route-map cache entries using 0 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
BGP using 19224 total bytes of memory
BGP activity 3023/2806 prefixes, 4932/4645 paths, scan interval 60 secs
302 networks peaked at 16:09:29 Sep 13 2022 UTC (1d02h ago)

Neighbor        V           AS MsgRcvd MsgSent   TblVer  InQ OutQ Up/Down  
State/PfxRcd
10.72.4.1       4        65001       8      10     1969    0    0 00:02:17        4
```

CASINO_VN:

```
C9500-Fusion#show ip bgp vpnv4 vrf CASINO_VN summary
BGP router identifier 10.72.192.1, local AS number 65000
BGP table version is 1969, main routing table version 1969
26 network entries using 6656 bytes of memory
36 path entries using 4896 bytes of memory
10 multipath network entries and 20 multipath paths
25/24 BGP path/bestpath attribute entries using 7800 bytes of memory
2 BGP AS-PATH entries using 48 bytes of memory
9 BGP extended community entries using 216 bytes of memory
0 BGP route-map cache entries using 0 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
BGP using 19616 total bytes of memory
BGP activity 3023/2806 prefixes, 4932/4645 paths, scan interval 60 secs
302 networks peaked at 16:09:29 Sep 13 2022 UTC (1d02h ago)

Neighbor        V           AS MsgRcvd MsgSent   TblVer  InQ OutQ Up/Down  
State/PfxRcd
10.72.4.33      4        65001      11      19     1969    0    0 00:03:40        5
```

CORP_VN:

```
C9500-Fusion#show ip bgp vpnv4 vrf CORP_VN summary
BGP router identifier 10.72.192.1, local AS number 65000
BGP table version is 1969, main routing table version 1969
30 network entries using 7680 bytes of memory
40 path entries using 5440 bytes of memory
10 multipath network entries and 20 multipath paths
25/24 BGP path/bestpath attribute entries using 7800 bytes of memory
2 BGP AS-PATH entries using 48 bytes of memory
9 BGP extended community entries using 216 bytes of memory
0 BGP route-map cache entries using 0 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
```

```
BGP using 21184 total bytes of memory
BGP activity 3023/2806 prefixes, 4932/4645 paths, scan interval 60 secs
302 networks peaked at 16:09:29 Sep 13 2022 UTC (1d02h ago)

Neighbor        V           AS MsgRcvd MsgSent   TblVer  InQ OutQ Up/Down
State/PfxRcd
10.72.4.5       4        65001      10      12     1969    0    0 00:04:12        8
```

IPTV_VN:

```
C9500-Fusion#show ip bgp vpnv4 vrf IPTV_VN summary
BGP router identifier 10.72.192.1, local AS number 65000
BGP table version is 1969, main routing table version 1969
26 network entries using 6656 bytes of memory
36 path entries using 4896 bytes of memory
10 multipath network entries and 20 multipath paths
25/24 BGP path/bestpath attribute entries using 7800 bytes of memory
2 BGP AS-PATH entries using 48 bytes of memory
9 BGP extended community entries using 216 bytes of memory
0 BGP route-map cache entries using 0 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
BGP using 19616 total bytes of memory
BGP activity 3023/2806 prefixes, 4932/4645 paths, scan interval 60 secs
302 networks peaked at 16:09:29 Sep 13 2022 UTC (1d02h ago)

Neighbor        V           AS MsgRcvd MsgSent   TblVer  InQ OutQ Up/Down
State/PfxRcd
10.72.4.17      4        65001      13      20     1969    0    0 00:05:26        5
```

PHONE_VN:

```
C9500-Fusion#show ip bgp vpnv4 vrf PHONE_VN summary
BGP router identifier 10.72.192.1, local AS number 65000
BGP table version is 1969, main routing table version 1969
24 network entries using 6144 bytes of memory
34 path entries using 4624 bytes of memory
10 multipath network entries and 20 multipath paths
25/24 BGP path/bestpath attribute entries using 7800 bytes of memory
2 BGP AS-PATH entries using 48 bytes of memory
9 BGP extended community entries using 216 bytes of memory
0 BGP route-map cache entries using 0 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
BGP using 18832 total bytes of memory
BGP activity 3023/2806 prefixes, 4932/4645 paths, scan interval 60 secs
302 networks peaked at 16:09:29 Sep 13 2022 UTC (1d02h ago)

Neighbor        V           AS MsgRcvd MsgSent   TblVer  InQ OutQ Up/Down
State/PfxRcd
10.72.4.25      4        65001      12      14     1969    0    0 00:06:07        3
```

POS_VN:

```
C9500-Fusion#show ip bgp vpnv4 vrf POS_VN summary
BGP router identifier 10.72.192.1, local AS number 65000
BGP table version is 1969, main routing table version 1969
23 network entries using 5888 bytes of memory
33 path entries using 4488 bytes of memory
10 multipath network entries and 20 multipath paths
25/24 BGP path/bestpath attribute entries using 7800 bytes of memory
2 BGP AS-PATH entries using 48 bytes of memory
9 BGP extended community entries using 216 bytes of memory
0 BGP route-map cache entries using 0 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
BGP using 18440 total bytes of memory
```

```
BGP activity 3023/2806 prefixes, 4932/4645 paths, scan interval 60 secs
302 networks peaked at 16:09:29 Sep 13 2022 UTC (1d02h ago)

Neighbor        V           AS MsgRcvd MsgSent   TblVer  InQ OutQ Up/Down
State/PfxRcd
10.72.4.29      4        65001      13      15     1969    0    0 00:06:45        2
```

3. Review GRT routing table to confirm all required subnets are being exported from each VRF:

> Two paths must be installed in the routing table for each subnet (going towards ACI).

```
C9500-Fusion#show ip route bgp
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, m - OMP
       n - NAT, Ni - NAT inside, No - NAT outside, Nd - NAT DIA
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       H - NHRP, G - NHRP registered, g - NHRP registration summary
       o - ODR, P - periodic downloaded static route, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR
       & - replicated local route overrides by connected

Gateway of last resort is 10.72.254.10 to network 0.0.0.0

      3.0.0.0/24 is subnetted, 1 subnets
B        3.3.3.0 [20/0] via 10.72.4.17 (IPTV_VN), 00:09:24
      10.0.0.0/8 is variably subnetted, 35 subnets, 8 masks
B        10.1.0.0/20 [20/0] via 10.72.4.13, 2d01h
B        10.72.0.0/24 [20/0] via 10.72.193.6, 2d01h
                      [20/0] via 10.72.193.2, 2d01h
B        10.72.1.0/24 [20/0] via 10.72.4.13, 2d01h
B        10.72.4.0/30 [20/0] via 10.72.4.1 (ACCESS_CTRL_VN), 00:08:51
B        10.72.4.4/30 [20/0] via 10.72.4.5 (CORP_VN), 00:09:16
B        10.72.4.16/30 [20/0] via 10.72.4.17 (IPTV_VN), 00:09:35
B        10.72.4.24/30 [20/0] via 10.72.4.25 (PHONE_VN), 00:09:05
B        10.72.4.28/30 [20/0] via 10.72.4.29 (POS_VN), 00:09:11
B        10.72.4.32/30 [20/0] via 10.72.4.33 (CASINO_VN), 00:09:37
B        10.72.4.36/30 is directly connected, 00:10:12, Vlan3010
B        10.72.6.0/23 [20/0] via 10.72.4.5 (CORP_VN), 00:09:16
B        10.72.8.0/24 [20/0] via 10.72.193.6, 2d01h
                      [20/0] via 10.72.193.2, 2d01h
B        10.72.9.0/24 [20/0] via 10.72.4.5 (CORP_VN), 00:09:16
B        10.72.10.0/23 [20/0] via 10.72.4.5 (CORP_VN), 00:09:16
B        10.72.49.0/24 [20/0] via 10.72.193.6, 2d01h
                       [20/0] via 10.72.193.2, 2d01h
B        10.72.50.0/24 [20/0] via 10.72.193.6, 2d01h
                       [20/0] via 10.72.193.2, 2d01h
B        10.72.51.0/24 [20/0] via 10.72.193.6, 2d01h
                       [20/0] via 10.72.193.2, 2d01h
B        10.72.56.0/24 [20/0] via 10.72.4.5 (CORP_VN), 00:09:16
B        10.72.124.0/22 [20/0] via 10.72.4.5 (CORP_VN), 00:09:16
B        10.72.194.0/26 [20/0] via 10.72.4.17 (IPTV_VN), 00:09:24
B        10.72.194.64/26 [20/0] via 10.72.4.5 (CORP_VN), 00:09:16
      172.16.0.0/16 is variably subnetted, 3 subnets, 3 masks
B        172.16.0.0/19 [20/0] via 10.72.4.25 (PHONE_VN), 00:09:05
B        172.16.150.0/24 [20/0] via 10.72.193.6, 2d01h
                         [20/0] via 10.72.193.2, 2d01h
B        172.16.176.0/21 [20/0] via 10.72.4.25 (PHONE_VN), 00:09:05
      172.18.0.0/16 is variably subnetted, 3 subnets, 2 masks
B        172.18.0.0/24 [20/0] via 10.72.4.33 (CASINO_VN), 00:09:37
```

```
B        172.18.1.0/24 [20/0] via 10.72.4.33 (CASINO_VN), 00:09:37
B        172.18.2.0/23 [20/0] via 10.72.4.33 (CASINO_VN), 00:09:37
      172.20.0.0/22 is subnetted, 2 subnets
B        172.20.0.0 [20/0] via 10.72.4.1 (ACCESS_CTRL_VN), 00:08:51
B        172.20.4.0 [20/0] via 10.72.4.1 (ACCESS_CTRL_VN), 00:08:51
      172.21.0.0/16 is variably subnetted, 5 subnets, 2 masks
B        172.21.40.0/21 [20/0] via 10.72.4.17 (IPTV_VN), 00:09:24
B        172.21.200.0/24 [20/0] via 10.72.193.6, 2d01h
                         [20/0] via 10.72.193.2, 2d01h
B        172.21.202.0/24 [20/0] via 10.72.193.6, 2d01h
                         [20/0] via 10.72.193.2, 2d01h
B        172.21.204.0/24 [20/0] via 10.72.193.6, 2d01h
                         [20/0] via 10.72.193.2, 2d01h
B        172.21.206.0/24 [20/0] via 10.72.193.6, 2d01h
                         [20/0] via 10.72.193.2, 2d01h
      172.22.0.0/19 is subnetted, 1 subnets
B        172.22.0.0 [20/0] via 10.72.4.17 (IPTV_VN), 00:09:24
      172.24.0.0/20 is subnetted, 1 subnets
B        172.24.0.0 [20/0] via 10.72.4.13, 2d01h
      172.25.0.0/21 is subnetted, 1 subnets
B        172.25.0.0 [20/0] via 10.72.4.1 (ACCESS_CTRL_VN), 00:08:51
      172.30.0.0/23 is subnetted, 1 subnets
B        172.30.38.0 [20/0] via 10.72.4.5 (CORP_VN), 00:09:16
B     192.168.6.0/24 [20/0] via 10.72.4.33 (CASINO_VN), 00:09:37
B     192.168.12.0/22 [20/0] via 10.72.4.29 (POS_VN), 00:09:11
```

4. Review the routing table for each VRF to confirm all required subnets are being imported.

Two paths must be installed in the routing table for each subnet (going towards ACI).

ACCESS_CTRL_VN:

```
C9500-Fusion#show ip route vrf ACCESS_CTRL_VN

Routing Table: ACCESS_CTRL_VN
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, m - OMP
       n - NAT, Ni - NAT inside, No - NAT outside, Nd - NAT DIA
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       H - NHRP, G - NHRP registered, g - NHRP registration summary
       o - ODR, P - periodic downloaded static route, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR
       & - replicated local route overrides by connected

Gateway of last resort is not set

      10.0.0.0/8 is variably subnetted, 22 subnets, 5 masks
B        10.1.0.0/20 [20/0] via 10.72.4.13, 00:12:12
B        10.72.0.0/24 [20/0] via 10.72.193.6, 00:12:12
                      [20/0] via 10.72.193.2, 00:12:12
B        10.72.1.0/24 [20/0] via 10.72.4.13, 00:12:12
C        10.72.4.0/30 is directly connected, Vlan3001
L        10.72.4.2/32 is directly connected, Vlan3001
B        10.72.4.12/30 is directly connected, 00:12:12, Vlan3004
L        10.72.4.14/32 is directly connected, Vlan3004
B        10.72.8.0/24 [20/0] via 10.72.193.6, 00:12:12
                      [20/0] via 10.72.193.2, 00:12:12
B        10.72.49.0/24 [20/0] via 10.72.193.6, 00:12:12
                       [20/0] via 10.72.193.2, 00:12:12
B        10.72.50.0/24 [20/0] via 10.72.193.6, 00:12:12
```

```
                           [20/0] via 10.72.193.2, 00:12:12
B         10.72.51.0/24 [20/0] via 10.72.193.6, 00:12:12
                           [20/0] via 10.72.193.2, 00:12:12
B         10.72.192.1/32 is directly connected, 00:12:12, Loopback0
B         10.72.193.0/30 is directly connected, 00:12:12, Port-channel110
L         10.72.193.1/32 is directly connected, Port-channel110
B         10.72.193.4/30 is directly connected, 00:12:12, Port-channel120
L         10.72.193.5/32 is directly connected, Port-channel120
B         10.72.193.8/30
            is directly connected, 00:12:12, FortyGigabitEthernet1/0/21
L         10.72.193.9/32 is directly connected, FortyGigabitEthernet1/0/21
B         10.72.254.8/29 is directly connected, 00:12:12, Vlan900
L         10.72.254.9/32 is directly connected, Vlan900
B         10.72.254.16/29 is directly connected, 00:12:12, Vlan696
L         10.72.254.17/32 is directly connected, Vlan696
      172.16.0.0/24 is subnetted, 1 subnets
B        172.16.150.0 [20/0] via 10.72.193.6, 00:12:12
                      [20/0] via 10.72.193.2, 00:12:12
      172.20.0.0/22 is subnetted, 2 subnets
B        172.20.0.0 [20/0] via 10.72.4.1, 00:10:51
B        172.20.4.0 [20/0] via 10.72.4.1, 00:10:51
      172.21.0.0/24 is subnetted, 4 subnets
B        172.21.200.0 [20/0] via 10.72.193.6, 00:12:12
                      [20/0] via 10.72.193.2, 00:12:12
B        172.21.202.0 [20/0] via 10.72.193.6, 00:12:12
                      [20/0] via 10.72.193.2, 00:12:12
B        172.21.204.0 [20/0] via 10.72.193.6, 00:12:12
                      [20/0] via 10.72.193.2, 00:12:12
B        172.21.206.0 [20/0] via 10.72.193.6, 00:12:12
                      [20/0] via 10.72.193.2, 00:12:12
      172.24.0.0/20 is subnetted, 1 subnets
B        172.24.0.0 [20/0] via 10.72.4.13, 00:12:12
      172.25.0.0/21 is subnetted, 1 subnets
B        172.25.0.0 [20/0] via 10.72.4.1, 00:10:51
      172.31.0.0/20 is subnetted, 1 subnets
B        172.31.0.0 [20/0] via 10.72.193.10, 00:12:12
```

CASINO_VN:

```
C9500-Fusion#show ip route vrf CASINO_VN

Routing Table: CASINO_VN
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, m - OMP
       n - NAT, Ni - NAT inside, No - NAT outside, Nd - NAT DIA
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       H - NHRP, G - NHRP registered, g - NHRP registration summary
       o - ODR, P - periodic downloaded static route, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR
       & - replicated local route overrides by connected

Gateway of last resort is not set

      10.0.0.0/8 is variably subnetted, 22 subnets, 5 masks
B        10.1.0.0/20 [20/0] via 10.72.4.13, 00:14:45
B        10.72.0.0/24 [20/0] via 10.72.193.6, 00:14:45
                      [20/0] via 10.72.193.2, 00:14:45
B        10.72.1.0/24 [20/0] via 10.72.4.13, 00:14:45
B        10.72.4.12/30 is directly connected, 00:14:45, Vlan3004
L        10.72.4.14/32 is directly connected, Vlan3004
C        10.72.4.32/30 is directly connected, Vlan3009
L        10.72.4.34/32 is directly connected, Vlan3009
```

```
B        10.72.8.0/24 [20/0] via 10.72.193.6, 00:14:45
                      [20/0] via 10.72.193.2, 00:14:45
B        10.72.49.0/24 [20/0] via 10.72.193.6, 00:14:45
                       [20/0] via 10.72.193.2, 00:14:45
B        10.72.50.0/24 [20/0] via 10.72.193.6, 00:14:45
                       [20/0] via 10.72.193.2, 00:14:45
B        10.72.51.0/24 [20/0] via 10.72.193.6, 00:14:45
                       [20/0] via 10.72.193.2, 00:14:45
B        10.72.192.1/32 is directly connected, 00:14:45, Loopback0
B        10.72.193.0/30 is directly connected, 00:14:45, Port-channel110
L        10.72.193.1/32 is directly connected, Port-channel110
B        10.72.193.4/30 is directly connected, 00:14:45, Port-channel120
L        10.72.193.5/32 is directly connected, Port-channel120
B        10.72.193.8/30
           is directly connected, 00:14:45, FortyGigabitEthernet1/0/21
L        10.72.193.9/32 is directly connected, FortyGigabitEthernet1/0/21
B        10.72.254.8/29 is directly connected, 00:14:45, Vlan900
L        10.72.254.9/32 is directly connected, Vlan900
B        10.72.254.16/29 is directly connected, 00:14:45, Vlan696
L        10.72.254.17/32 is directly connected, Vlan696
      172.16.0.0/24 is subnetted, 1 subnets
B        172.16.150.0 [20/0] via 10.72.193.6, 00:14:45
                      [20/0] via 10.72.193.2, 00:14:45
      172.18.0.0/16 is variably subnetted, 3 subnets, 2 masks
B        172.18.0.0/24 [20/0] via 10.72.4.33, 00:13:57
B        172.18.1.0/24 [20/0] via 10.72.4.33, 00:13:57
B        172.18.2.0/23 [20/0] via 10.72.4.33, 00:13:57
      172.21.0.0/24 is subnetted, 4 subnets
B        172.21.200.0 [20/0] via 10.72.193.6, 00:14:45
                      [20/0] via 10.72.193.2, 00:14:45
B        172.21.202.0 [20/0] via 10.72.193.6, 00:14:45
                      [20/0] via 10.72.193.2, 00:14:45
B        172.21.204.0 [20/0] via 10.72.193.6, 00:14:45
                      [20/0] via 10.72.193.2, 00:14:45
B        172.21.206.0 [20/0] via 10.72.193.6, 00:14:45
                      [20/0] via 10.72.193.2, 00:14:45
      172.24.0.0/20 is subnetted, 1 subnets
B        172.24.0.0 [20/0] via 10.72.4.13, 00:14:45
      172.31.0.0/20 is subnetted, 1 subnets
B        172.31.0.0 [20/0] via 10.72.193.10, 00:14:45
B     192.168.6.0/24 [20/0] via 10.72.4.33, 00:13:57
```

CORP_VN:

```
C9500-Fusion#show ip route vrf CORP_VN

Routing Table: CORP_VN
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, m - OMP
       n - NAT, Ni - NAT inside, No - NAT outside, Nd - NAT DIA
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       H - NHRP, G - NHRP registered, g - NHRP registration summary
       o - ODR, P - periodic downloaded static route, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR
       & - replicated local route overrides by connected

Gateway of last resort is 10.72.254.10 to network 0.0.0.0

B*    0.0.0.0/0 [20/0] via 10.72.254.10, 00:15:18
      10.0.0.0/8 is variably subnetted, 28 subnets, 8 masks
B        10.1.0.0/20 [20/0] via 10.72.4.13, 00:15:18
```

```
B        10.72.0.0/24 [20/0] via 10.72.193.6, 00:15:18
                      [20/0] via 10.72.193.2, 00:15:18
B        10.72.1.0/24 [20/0] via 10.72.4.13, 00:15:18
C        10.72.4.4/30 is directly connected, Vlan3002
L        10.72.4.6/32 is directly connected, Vlan3002
B        10.72.4.12/30 is directly connected, 00:15:18, Vlan3004
L        10.72.4.14/32 is directly connected, Vlan3004
B        10.72.6.0/23 [20/0] via 10.72.4.5, 00:14:22
B        10.72.8.0/24 [20/0] via 10.72.193.6, 00:15:18
                      [20/0] via 10.72.193.2, 00:15:18
B        10.72.9.0/24 [20/0] via 10.72.4.5, 00:14:22
B        10.72.10.0/23 [20/0] via 10.72.4.5, 00:14:22
B        10.72.49.0/24 [20/0] via 10.72.193.6, 00:15:18
                       [20/0] via 10.72.193.2, 00:15:18
B        10.72.50.0/24 [20/0] via 10.72.193.6, 00:15:18
                       [20/0] via 10.72.193.2, 00:15:18
B        10.72.51.0/24 [20/0] via 10.72.193.6, 00:15:18
                       [20/0] via 10.72.193.2, 00:15:18
B        10.72.56.0/24 [20/0] via 10.72.4.5, 00:14:22
B        10.72.124.0/22 [20/0] via 10.72.4.5, 00:14:22
B        10.72.192.1/32 is directly connected, 00:15:18, Loopback0
B        10.72.193.0/30 is directly connected, 00:15:18, Port-channel110
L        10.72.193.1/32 is directly connected, Port-channel110
B        10.72.193.4/30 is directly connected, 00:15:18, Port-channel120
L        10.72.193.5/32 is directly connected, Port-channel120
B        10.72.193.8/30
           is directly connected, 00:15:18, FortyGigabitEthernet1/0/21
L        10.72.193.9/32 is directly connected, FortyGigabitEthernet1/0/21
B        10.72.194.64/26 [20/0] via 10.72.4.5, 00:14:22
B        10.72.254.8/29 is directly connected, 00:15:18, Vlan900
L        10.72.254.9/32 is directly connected, Vlan900
B        10.72.254.16/29 is directly connected, 00:15:18, Vlan696
L        10.72.254.17/32 is directly connected, Vlan696
      172.16.0.0/24 is subnetted, 1 subnets
B        172.16.150.0 [20/0] via 10.72.193.6, 00:15:18
                      [20/0] via 10.72.193.2, 00:15:18
      172.21.0.0/24 is subnetted, 4 subnets
B        172.21.200.0 [20/0] via 10.72.193.6, 00:15:18
                      [20/0] via 10.72.193.2, 00:15:18
B        172.21.202.0 [20/0] via 10.72.193.6, 00:15:18
                      [20/0] via 10.72.193.2, 00:15:18
B        172.21.204.0 [20/0] via 10.72.193.6, 00:15:18
                      [20/0] via 10.72.193.2, 00:15:18
B        172.21.206.0 [20/0] via 10.72.193.6, 00:15:18
                      [20/0] via 10.72.193.2, 00:15:18
      172.24.0.0/20 is subnetted, 1 subnets
B        172.24.0.0 [20/0] via 10.72.4.13, 00:15:18
      172.30.0.0/23 is subnetted, 1 subnets
B        172.30.38.0 [20/0] via 10.72.4.5, 00:14:22
      172.31.0.0/20 is subnetted, 1 subnets
B        172.31.0.0 [20/0] via 10.72.193.10, 00:15:18
```

IPTV_VN:

```
C9500-Fusion#show ip route vrf IPTV_VN
Routing Table: IPTV_VN
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, m - OMP
       n - NAT, Ni - NAT inside, No - NAT outside, Nd - NAT DIA
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       H - NHRP, G - NHRP registered, g - NHRP registration summary
       o - ODR, P - periodic downloaded static route, l - LISP
       a - application route
```

```
       + - replicated route, % - next hop override, p - overrides from PfR
       & - replicated local route overrides by connected

Gateway of last resort is not set

      3.0.0.0/24 is subnetted, 1 subnets
B        3.3.3.0 [20/0] via 10.72.4.17, 00:15:15
      10.0.0.0/8 is variably subnetted, 23 subnets, 6 masks
B        10.1.0.0/20 [20/0] via 10.72.4.13, 00:16:03
B        10.72.0.0/24 [20/0] via 10.72.193.6, 00:16:03
                      [20/0] via 10.72.193.2, 00:16:03
B        10.72.1.0/24 [20/0] via 10.72.4.13, 00:16:03
B        10.72.4.12/30 is directly connected, 00:16:03, Vlan3004
L        10.72.4.14/32 is directly connected, Vlan3004
C        10.72.4.16/30 is directly connected, Vlan3005
L        10.72.4.18/32 is directly connected, Vlan3005
B        10.72.8.0/24 [20/0] via 10.72.193.6, 00:16:03
                      [20/0] via 10.72.193.2, 00:16:03
B        10.72.49.0/24 [20/0] via 10.72.193.6, 00:16:03
                       [20/0] via 10.72.193.2, 00:16:03
B        10.72.50.0/24 [20/0] via 10.72.193.6, 00:16:03
                       [20/0] via 10.72.193.2, 00:16:03
B        10.72.51.0/24 [20/0] via 10.72.193.6, 00:16:03
                       [20/0] via 10.72.193.2, 00:16:03
B        10.72.192.1/32 is directly connected, 00:16:03, Loopback0
B        10.72.193.0/30 is directly connected, 00:16:03, Port-channel110
L        10.72.193.1/32 is directly connected, Port-channel110
B        10.72.193.4/30 is directly connected, 00:16:03, Port-channel120
L        10.72.193.5/32 is directly connected, Port-channel120
B        10.72.193.8/30
           is directly connected, 00:16:03, FortyGigabitEthernet1/0/21
L        10.72.193.9/32 is directly connected, FortyGigabitEthernet1/0/21
B        10.72.194.0/26 [20/0] via 10.72.4.17, 00:15:15
B        10.72.254.8/29 is directly connected, 00:16:03, Vlan900
L        10.72.254.9/32 is directly connected, Vlan900
B        10.72.254.16/29 is directly connected, 00:16:03, Vlan696
L        10.72.254.17/32 is directly connected, Vlan696
      172.16.0.0/24 is subnetted, 1 subnets
B        172.16.150.0 [20/0] via 10.72.193.6, 00:16:03
                      [20/0] via 10.72.193.2, 00:16:03
      172.21.0.0/16 is variably subnetted, 5 subnets, 2 masks
B        172.21.40.0/21 [20/0] via 10.72.4.17, 00:15:15
B        172.21.200.0/24 [20/0] via 10.72.193.6, 00:16:03
                         [20/0] via 10.72.193.2, 00:16:03
B        172.21.202.0/24 [20/0] via 10.72.193.6, 00:16:03
                         [20/0] via 10.72.193.2, 00:16:03
B        172.21.204.0/24 [20/0] via 10.72.193.6, 00:16:03
                         [20/0] via 10.72.193.2, 00:16:03
B        172.21.206.0/24 [20/0] via 10.72.193.6, 00:16:03
                         [20/0] via 10.72.193.2, 00:16:03
      172.22.0.0/19 is subnetted, 1 subnets
B        172.22.0.0 [20/0] via 10.72.4.17, 00:15:15
      172.24.0.0/20 is subnetted, 1 subnets
B        172.24.0.0 [20/0] via 10.72.4.13, 00:16:03
      172.31.0.0/20 is subnetted, 1 subnets
B        172.31.0.0 [20/0] via 10.72.193.10, 00:16:03
```

PHONE_VN:

```
C9500-Fusion#show ip route vrf PHONE_VN

Routing Table: PHONE_VN
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, m - OMP
```

```
       n - NAT, Ni - NAT inside, No - NAT outside, Nd - NAT DIA
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       H - NHRP, G - NHRP registered, g - NHRP registration summary
       o - ODR, P - periodic downloaded static route, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR
       & - replicated local route overrides by connected

Gateway of last resort is not set

      10.0.0.0/8 is variably subnetted, 22 subnets, 5 masks
B        10.1.0.0/20 [20/0] via 10.72.4.13, 00:16:34
B        10.72.0.0/24 [20/0] via 10.72.193.6, 00:16:34
                      [20/0] via 10.72.193.2, 00:16:34
B        10.72.1.0/24 [20/0] via 10.72.4.13, 00:16:34
B        10.72.4.12/30 is directly connected, 00:16:34, Vlan3004
L        10.72.4.14/32 is directly connected, Vlan3004
C        10.72.4.24/30 is directly connected, Vlan3007
L        10.72.4.26/32 is directly connected, Vlan3007
B        10.72.8.0/24 [20/0] via 10.72.193.6, 00:16:34
                      [20/0] via 10.72.193.2, 00:16:34
B        10.72.49.0/24 [20/0] via 10.72.193.6, 00:16:34
                       [20/0] via 10.72.193.2, 00:16:34
B        10.72.50.0/24 [20/0] via 10.72.193.6, 00:16:34
                       [20/0] via 10.72.193.2, 00:16:34
B        10.72.51.0/24 [20/0] via 10.72.193.6, 00:16:34
                       [20/0] via 10.72.193.2, 00:16:34
B        10.72.192.1/32 is directly connected, 00:16:34, Loopback0
B        10.72.193.0/30 is directly connected, 00:16:34, Port-channel110
L        10.72.193.1/32 is directly connected, Port-channel110
B        10.72.193.4/30 is directly connected, 00:16:34, Port-channel120
L        10.72.193.5/32 is directly connected, Port-channel120
B        10.72.193.8/30
           is directly connected, 00:16:34, FortyGigabitEthernet1/0/21
L        10.72.193.9/32 is directly connected, FortyGigabitEthernet1/0/21
B        10.72.254.8/29 is directly connected, 00:16:34, Vlan900
L        10.72.254.9/32 is directly connected, Vlan900
B        10.72.254.16/29 is directly connected, 00:16:34, Vlan696
L        10.72.254.17/32 is directly connected, Vlan696
      172.16.0.0/16 is variably subnetted, 3 subnets, 3 masks
B        172.16.0.0/19 [20/0] via 10.72.4.25, 00:15:27
B        172.16.150.0/24 [20/0] via 10.72.193.6, 00:16:34
                         [20/0] via 10.72.193.2, 00:16:34
B        172.16.176.0/21 [20/0] via 10.72.4.25, 00:15:27
      172.21.0.0/24 is subnetted, 4 subnets
B        172.21.200.0 [20/0] via 10.72.193.6, 00:16:34
                      [20/0] via 10.72.193.2, 00:16:34
B        172.21.202.0 [20/0] via 10.72.193.6, 00:16:34
                      [20/0] via 10.72.193.2, 00:16:34
B        172.21.204.0 [20/0] via 10.72.193.6, 00:16:34
                      [20/0] via 10.72.193.2, 00:16:34
B        172.21.206.0 [20/0] via 10.72.193.6, 00:16:34
                      [20/0] via 10.72.193.2, 00:16:34
      172.24.0.0/20 is subnetted, 1 subnets
B        172.24.0.0 [20/0] via 10.72.4.13, 00:16:34
      172.31.0.0/20 is subnetted, 1 subnets
B        172.31.0.0 [20/0] via 10.72.193.10, 00:16:34
```

POS_VN:

```
C9500-Fusion#sh ip route vrf POS_VN
Routing Table: POS_VN
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
```

```
       E1 - OSPF external type 1, E2 - OSPF external type 2, m - OMP
       n - NAT, Ni - NAT inside, No - NAT outside, Nd - NAT DIA
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       H - NHRP, G - NHRP registered, g - NHRP registration summary
       o - ODR, P - periodic downloaded static route, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR
       & - replicated local route overrides by connected

Gateway of last resort is not set

      10.0.0.0/8 is variably subnetted, 22 subnets, 5 masks
B        10.1.0.0/20 [20/0] via 10.72.4.13, 00:18:09
B        10.72.0.0/24 [20/0] via 10.72.193.6, 00:18:09
                      [20/0] via 10.72.193.2, 00:18:09
B        10.72.1.0/24 [20/0] via 10.72.4.13, 00:18:09
B        10.72.4.12/30 is directly connected, 00:18:09, Vlan3004
L        10.72.4.14/32 is directly connected, Vlan3004
C        10.72.4.28/30 is directly connected, Vlan3008
L        10.72.4.30/32 is directly connected, Vlan3008
B        10.72.8.0/24 [20/0] via 10.72.193.6, 00:18:09
                      [20/0] via 10.72.193.2, 00:18:09
B        10.72.49.0/24 [20/0] via 10.72.193.6, 00:18:09
                       [20/0] via 10.72.193.2, 00:18:09
B        10.72.50.0/24 [20/0] via 10.72.193.6, 00:18:09
                       [20/0] via 10.72.193.2, 00:18:09
B        10.72.51.0/24 [20/0] via 10.72.193.6, 00:18:09
                       [20/0] via 10.72.193.2, 00:18:09
B        10.72.192.1/32 is directly connected, 00:18:09, Loopback0
B        10.72.193.0/30 is directly connected, 00:18:09, Port-channel110
L        10.72.193.1/32 is directly connected, Port-channel110
B        10.72.193.4/30 is directly connected, 00:18:09, Port-channel120
L        10.72.193.5/32 is directly connected, Port-channel120
B        10.72.193.8/30
           is directly connected, 00:18:09, FortyGigabitEthernet1/0/21
L        10.72.193.9/32 is directly connected, FortyGigabitEthernet1/0/21
B        10.72.254.8/29 is directly connected, 00:18:09, Vlan900
L        10.72.254.9/32 is directly connected, Vlan900
B        10.72.254.16/29 is directly connected, 00:18:09, Vlan696
L        10.72.254.17/32 is directly connected, Vlan696
      172.16.0.0/24 is subnetted, 1 subnets
B        172.16.150.0 [20/0] via 10.72.193.6, 00:18:09
                      [20/0] via 10.72.193.2, 00:18:09
      172.21.0.0/24 is subnetted, 4 subnets
B        172.21.200.0 [20/0] via 10.72.193.6, 00:18:09
                      [20/0] via 10.72.193.2, 00:18:09
B        172.21.202.0 [20/0] via 10.72.193.6, 00:18:09
                      [20/0] via 10.72.193.2, 00:18:09
B        172.21.204.0 [20/0] via 10.72.193.6, 00:18:09
                      [20/0] via 10.72.193.2, 00:18:09
B        172.21.206.0 [20/0] via 10.72.193.6, 00:18:09
                      [20/0] via 10.72.193.2, 00:18:09
      172.24.0.0/20 is subnetted, 1 subnets
B        172.24.0.0 [20/0] via 10.72.4.13, 00:18:09
      172.31.0.0/20 is subnetted, 1 subnets
B        172.31.0.0 [20/0] via 10.72.193.10, 00:18:09
B     192.168.12.0/22 [20/0] via 10.72.4.29, 00:17:08
```

5. Execute tests to confirm macro-segmentation is working as expected. For example, CASINO_VN cannot talk to CORP_VN by default:

```
C9500-Fusion#ping vrf CASINO_VN 10.72.4.6
Type escape sequence to abort.
```

```
Sending 5, 100-byte ICMP Echos to 10.72.4.6, timeout is 2 seconds:
.....
Success rate is 0 percent (0/5)
```

6. Ask NCL to execute health check validations for the end points on the network to confirm that no traffic is impacted, or any issues arise.
7. If all checks pass, save the changes to the running-configuration for both Fusion and Border.

```
write memory
```

8. Save a backup of the configuration locally on the Fusion Router and Border:

```
copy running-configuration bootflash:post-macro-segmentation-changes.txt
```

3 Adding inline tagging to the Border and the Fusion nodes

3.1 Backup config

1. Backup the running config on both the border node and the fusion node.(This might have been completed in the previous steps)
2. Verify the current state of the Port Channel

```
sh etherchannel 100 sum
```

3. Verify that the fusion has the CTS configuration, PAC, environment data,

```
sh cts pac
sh cts env
```

4. Verify cts enforcement configuration is in place on both Border and Fusion, if missing, we fix the vlans:

```
Show vlan
cts role-based enforcement
cts role-based enforcement vlan 3001-3009
```

5. Verify cts role-based sgt-mal vrf CORPORATE_VN all on both the Border node and the Fusion node. Make sure SGT 2 is present

```
sh cts role-based sgt-map vrf COPRORATE_VN all
```

3.2 Changes

1. There are four physical links in the port channel 100, and we will remove two links on both the border switch and the fusion switch:
   Border Switch

```
int range for2/5/0/24,fo1/6/0/24
No channel-group 100 mode active
```

   Fusion Router

```
int range Fo1/0/28-29
 No channel-group 100 mode active
```

2. Add the cts commands to both switches, please note that the physical link will bounce

```
cts manual
propagate sgt
policy static sgt 2 trusted
exit
end
```

6. Verify the current state of the Port Channel

```
sh etherchannel 100 sum
```

7. We will remove two remaining links on both the border switch and the fusion switch from the port channel:
   Border Switch

```
int range for2/6/0/24,fo1/5/0/24
No channel-group 100 mode active
```

   Fusion Router

```
int range Fo2/0/28-29
 No channel-group 100 mode active
```

   Switch back to the original two interfaces and add them back to the port channel

```
int range for2/5/0/24,fo1/6/0/24
 channel-group 100 mode active
```

   Fusion Router

```
int range Fo1/0/28-29
 channel-group 100 mode active
```

8. Verify the port channel is up

```
sh etherchannel 100 sum
```

9. Break Point: Verify that the routing and end points are working before continuing.

10. We will join the two remaining links on both the border switch and the fusion switch to the port channel:
    Border Switch

```
int range for2/6/0/24,fo1/5/0/24
cts manual
propagate sgt
policy static sgt 2 trusted
exit
channel-group 100 mode active
end
```

    Fusion Router

```
int range Fo2/0/28-29
cts manual
propagate sgt
policy static sgt 2 trusted
exit
channel-group 100 mode active
end
```

11. Verify the port channel is up

```
sh etherchannel 100 sum
```

12. Verify that the routing and end points again.

3.3 Validation

1. Show channel-group 100 sum to display active membership of the ports
2. Execute all validations from section 3 again to confirm routing is working as expected after the changes.
3. If allowed, capture a traffic on the fusion from a client and we should be able to see the Cisco Meta Data including the SGT.
4. If all checks pass, save the changes to the running-configuration for both Fusion and Border.

```
write memory
```

5. Save a backup of the configuration locally on the Fusion Router and Border:

```
copy running-configuration bootflash:post-macro-segmentation-changes.txt
```

4 Back out and removal

4.1 Backout of the Inline-tagging changes

1. There are four physical links in the port channel 100, and we will remove links on both the border switch and the fusion switch:
   Border Switch

```
int range tw1/5/24,tw2/5/24,1/6/24,2/6/24
no channel-group 100 mode active
```

   Fusion Router

```
int range Fo1/0/28-29,Fo2/0/28-29
no channel-group 100 mode active
```

2. Remove the cts commands to both switches, please note that the physical link will bounce

```
no Cts manual
channel-group 100 mode active
end
```

13. Verify the current state of the Port Channel

```
sh etherchannel 100 sum
```

4.2 Routing changes for macro-segmentation

4.2.1 Delete VRFs

Enter the following to the Fusion Router:

```
!
no vrf definition ACCESS_CTRL_VN
no vrf definition CORP_VN
no vrf definition IPTV_VN
no vrf definition PHONE_VN
no vrf definition POS_VN
no vrf definition CASINO_VN
```

4.2.2 Add IP Addressing back to SVIs

Enter the following to the Fusion Router:

```
!
interface Vlan3001
ip address 10.72.4.2 255.255.255.252
!
interface Vlan3002
ip address 10.72.4.6 255.255.255.252
ip pim sparse-mode
!
interface Vlan3005
```

```
ip address 10.72.4.18 255.255.255.252
ip pim sparse-mode
!
interface Vlan3007
ip address 10.72.4.26 255.255.255.252
!
interface Vlan3008
ip address 10.72.4.30 255.255.255.252
!
interface Vlan3009
ip address 10.72.4.34 255.255.255.252
```

4.2.3 Add BGP peering through GRT

Enter the following to the Fusion Router:

```
!
router bgp 65000
neighbor 10.72.4.1 remote-as 65001
neighbor 10.72.4.5 remote-as 65001
neighbor 10.72.4.17 remote-as 65001
neighbor 10.72.4.25 remote-as 65001
neighbor 10.72.4.29 remote-as 65001
neighbor 10.72.4.33 remote-as 65001
```

4.2.4 Confirm routing

Compare the output of “show ip route” with the output taken in section 2.1 “Preparation” to confirm it matches what was configured before applying these changes. Ping tests are also recommended to confirm back-out.

4.2.5 Delete IP prefix-lists to import from GRT to each VN

Enter the following to the Fusion Router:

```
!
no ip prefix-list PERMIT-ALL-BUT-Q0
```

4.2.6 Delete Route-maps to export prefix-lists to GRT

Enter the following to the Fusion Router:

```
!
no route-map PHONE-TO-GRT permit 10
no route-map ACCESS_CTRL-TO-GRT permit 10
no route-map IPTV-TO-GRT permit 10
no route-map CASINO-TO-GRT permit 10
no route-map CORP-TO-GRT permit 10
no route-map POS-TO-GRT permit 10
```

4.2.7 Delete Route-maps to import prefix-lists from GRT

Enter the following to the Fusion Router:

```
!
no route-map IMPORT-TO-CORP permit 10
no route-map IMPORT-TO-IPTV permit 10
no route-map IMPORT-TO-PHONE permit 10
no route-map IMPORT-TO-CASINO permit 10
no route-map IMPORT-TO-POS permit 10
no route-map IMPORT-TO-ACCESS_CTRL permit 10
```

5 Appendix A – Macro-segmentation with specific subnets

5.1 Create VRFs

From a console connection to the Fusion Router, create the VRFs to match the existing VNs in the SD-Access Border:

```
vrf definition ACCESS_CTRL_VN
rd 1:4100
!
address-family ipv4
route-target export 1:4100
route-target import 1:4100
exit-address-family
!
vrf definition CORP_VN
rd 1:4105
!
address-family ipv4
route-target export 1:4105
route-target import 1:4105
exit-address-family
!
vrf definition IPTV_VN
rd 1:4099
!
address-family ipv4
route-target export 1:4099
route-target import 1:4099
exit-address-family
!
vrf definition PHONE_VN
rd 1:4102
!
address-family ipv4
route-target export 1:4102
route-target import 1:4102
exit-address-family
!
vrf definition POS_VN
rd 1:4101
!
address-family ipv4
route-target export 1:4101
route-target import 1:4101
exit-address-family
!
vrf definition CASINO_VN
rd 1:4107
!
address-family ipv4
route-target export 1:4107
route-target import 1:4107
exit-address-family
!
```

5.2 Create IP prefix-lists to export from each VRF to GRT

Enter the following to the Fusion Router:

```
ip prefix-list ACCESS_CTRL-TO-GRT seq 5 permit 172.25.0.0/21
ip prefix-list ACCESS_CTRL-TO-GRT seq 10 permit 172.20.0.0/22
ip prefix-list ACCESS_CTRL-TO-GRT seq 15 permit 172.20.4.0/22
ip prefix-list ACCESS_CTRL-TO-GRT seq 20 permit 172.20.8.0/22
ip prefix-list ACCESS_CTRL-TO-GRT seq 25 permit 172.20.12.0/22

ip prefix-list CASINO-TO-GRT seq 5 permit 172.18.2.0/23
ip prefix-list CASINO-TO-GRT seq 10 permit 172.18.0.0/24
ip prefix-list CASINO-TO-GRT seq 15 permit 192.168.6.0/24
ip prefix-list CASINO-TO-GRT seq 20 permit 172.18.1.0/24

ip prefix-list CORP-TO-GRT seq 5 permit 10.72.10.0/23
ip prefix-list CORP-TO-GRT seq 10 permit 10.72.6.0/23
ip prefix-list CORP-TO-GRT seq 15 permit 172.30.38.0/23
ip prefix-list CORP-TO-GRT seq 20 permit 10.72.124.0/22
ip prefix-list CORP-TO-GRT seq 25 permit 10.72.9.0/24
ip prefix-list CORP-TO-GRT seq 30 permit 10.72.56.0/24

ip prefix-list IPTV-TO-GRT seq 5 permit 172.21.0.0/19
ip prefix-list IPTV-TO-GRT seq 10 permit 172.21.40.0/21

ip prefix-list PHONE-TO-GRT seq 5 permit 172.16.176.0/21
ip prefix-list PHONE-TO-GRT seq 10 permit 172.16.0.0/19

ip prefix-list POS-TO-GRT seq 5 permit 192.168.12.0/22
!
```

5.3 Create IP prefix-lists to import from GRT to each VN

Enter the following to the Fusion Router:

```
ip prefix-list IMPORT-TO-ACCESS_CTRL seq 5 permit 10.72.49.0/24
ip prefix-list IMPORT-TO-ACCESS_CTRL seq 10 permit 10.72.50.0/24
ip prefix-list IMPORT-TO-ACCESS_CTRL seq 15 permit 10.72.51.0/24

ip prefix-list IMPORT-TO-CASINO seq 5 permit 10.72.8.0/24
ip prefix-list IMPORT-TO-CASINO seq 10 permit 10.72.49.0/24
ip prefix-list IMPORT-TO-CASINO seq 15 permit 10.72.50.0/24
ip prefix-list IMPORT-TO-CASINO seq 20 permit 10.72.51.0/24

ip prefix-list IMPORT-TO-CORP seq 5 permit 10.72.8.0/24
ip prefix-list IMPORT-TO-CORP seq 10 permit 10.72.49.0/24
ip prefix-list IMPORT-TO-CORP seq 15 permit 10.72.50.0/24
ip prefix-list IMPORT-TO-CORP seq 20 permit 10.72.51.0/24
ip prefix-list IMPORT-TO-CORP seq 25 permit 172.21.202.0/24
ip prefix-list IMPORT-TO-CORP seq 30 permit 10.72.0.0/24
ip prefix-list IMPORT-TO-CORP seq 35 permit 0.0.0.0/0

ip prefix-list IMPORT-TO-IPTV seq 5 permit 10.72.49.0/24
ip prefix-list IMPORT-TO-IPTV seq 10 permit 10.72.50.0/24
ip prefix-list IMPORT-TO-IPTV seq 15 permit 10.72.51.0/24
ip prefix-list IMPORT-TO-IPTV seq 20 permit 172.21.200.0/24
```

```
ip prefix-list IMPORT-TO-PHONE seq 5 permit 172.16.150.0/24

ip prefix-list IMPORT-TO-POS seq 5 permit 10.72.49.0/24
ip prefix-list IMPORT-TO-POS seq 10 permit 10.72.50.0/24
ip prefix-list IMPORT-TO-POS seq 15 permit 10.72.51.0/24
!
```

5.4 Create Route-maps to export prefix-lists to GRT

Enter the following to the Fusion Router:

```
route-map PHONE-TO-GRT permit 10
match ip address prefix-list PHONE-TO-GRT
!
route-map ACCESS_CTRL-TO-GRT permit 10
match ip address prefix-list ACCESS_CTRL-TO-GRT
!
route-map IPTV-TO-GRT permit 10
match ip address prefix-list IPTV-TO-GRT
!
route-map POS-TO-GRT permit 10
match ip address prefix-list POS-TO-GRT
!
route-map CORP-TO-GRT permit 10
match ip address prefix-list CORP-TO-GRT
!
route-map CASINO-TO-GRT permit 10
match ip address prefix-list CASINO-TO-GRT
!
```

5.5 Create Route-maps to import prefix-lists from GRT

Enter the following to the Fusion Router:

```
route-map IMPORT-TO-CASINO permit 10
match ip address prefix-list IMPORT-TO-CASINO
!
route-map IMPORT-TO-CORP permit 10
match ip address prefix-list IMPORT-TO-CORP
!
route-map IMPORT-TO-POS permit 10
match ip address prefix-list IMPORT-TO-POS
!
route-map IMPORT-TO-IPTV permit 10
match ip address prefix-list IMPORT-TO-IPTV
!
route-map IMPORT-TO-ACCESS_CTRL permit 10
match ip address prefix-list IMPORT-TO-ACCESS_CTRL
!
route-map IMPORT-TO-PHONE permit 10
match ip address prefix-list IMPORT-TO-PHONE
!
```

5.6 Associate required route-maps to each VRF

Enter the following to the Fusion Router:

```
vrf definition ACCESS_CTRL_VN
rd 1:4100
!
address-family ipv4
import ipv4 unicast map IMPORT-TO-ACCESS_CTRL
export ipv4 unicast map ACCESS_CTRL-TO-GRT
exit-address-family
!
vrf definition CORP_VN
rd 1:4105
!
address-family ipv4
import ipv4 unicast map IMPORT-TO-CORP
export ipv4 unicast map CORP-TO-GRT
exit-address-family
!
vrf definition IPTV_VN
rd 1:4099
!
address-family ipv4
import ipv4 unicast map IMPORT-TO-IPTV
export ipv4 unicast map IPTV-TO-GRT
exit-address-family
!
vrf definition PHONE_VN
rd 1:4102
!
address-family ipv4
import ipv4 unicast map IMPORT-TO-PHONE
export ipv4 unicast map PHONE-TO-GRT
exit-address-family
!
vrf definition POS_VN
rd 1:4101
!
address-family ipv4
import ipv4 unicast map IMPORT-TO-POS
export ipv4 unicast map POS-TO-GRT
exit-address-family
!
vrf definition CASINO_VN
rd 1:4107
!
address-family ipv4
import ipv4 unicast map IMPORT-TO-CASINO
export ipv4 unicast map CASINO-TO-GRT
exit-address-family
!
```

After this has been completed, proceed to execute the final steps from section 2.7 “BGP peering changes” to 2.12 “Validation”.

6 Appendix B – Inter-VRF leaking

In the future when NCL requires inter-VRF leaking, such leaking can be configured to allow traffic from certain VRFs to reach other VRFs through the Fusion Router.

For example, to allow communication from CORP_VN to IPTV_VN, configure the following:

From Fusion:

```
vrf definition CORP_VN
rd 1:4104
!
address-family ipv4
route-target import 1:4103
exit-address-family
!
vrf definition IPTV_VN
rd 1:4103
!
address-family ipv4
route-target import 1:4104
exit-address-family
!
```

From Border:

```
router bgp 65001
address-family ipv4 vrf CORP_VN
neighbor 10.72.4.6 allowas-in
exit-address-family
!
address-family ipv4 vrf IPTV_VN
neighbor 10.72.4.18 allowas-in
exit-address-family
!
```

7 Appendix C - Acronym Listing or Full Glossary

Table 2 Acronyms

Term	Definition
AAA	Authentication, Authorization, Accounting
ACL	Access Lists
ASR	Aggregation Services Router
ASA	Adaptive Security Appliance
BGP	Border Gateway Protocol
BPDU	Bridge Protocol Data Units
BSS	Business Support Systems
CAA	Clean Access Agent
CAM	Clean Access Manager
CAR	Committed Access Rate
CAS	Clean Access Server
CI	Configuration ITEM
CIC	CISCO Info Center
CLI	Command Line Interface
CM	Central Manager
CMDB	Configuration Management Database
CPE	Customer Premise Equipment
CPU	Central Processing Unit
cRTP	Compressed Real Time Protocol
CSM	CISCO Security Manager
CWCS	CISCO Wireless Control System
DHCP	Dynamic Host Control Protocol
DNS	Domain Name Server
DSU	Digital Subscriber Unit
FR	Frame Relay
FRAD	Frame Relay Access Device
FTP	File Transfer Protocol
FW	Firewall
GRE	Generic Routing Encapsulation
HDLC	High level Data Link Control
HLD	High Level Design
HSRP	Hot Standby Redundancy Protocol
HTTP	Hyper Text Transfer Protocol
HTTPS	Secure http
IB	In-Band
IDS	Intrusion Detection System
IOS	Internetworking Operating System
IP	Internet Protocol
IPeFR	IP enabled Frame Relay
IPS	Intrusion Protection System
IPSec	IP Secure

Term	Definition
ISE	Identity Services Engine
ISP	Internet Service Provider
ISR	Integrated Services Router
KEK	Key Encryption Key
KS	Key Server
LAN	Local Area Network
LLD	Low Level Design
LLQ	Low Latency Queuing
LWAPP	Light Weight Access Point
MAC	Media Access Control
MARS	CISCO Security Monitoring And Response System
MDNS	Managed Data Network Service
MPLS	Multiple Protocol Label Switching
MSE	Mobility Service Engine
MS	Managed Services
NAC	Network Access Control
NOC	Network Operation Center
NTP	Network Time Protocol
OER	Optimized Edge Routing
OOB	Out Of Band
OSPF	Open Shortest Path First
OSS	Operational Support Systems
OTV	Overlay Transport Virtualization
OWL	One Way Latency
PER	Provider Edge Router
PIMSM	Protocol Independent Multicast Sparse Mode
PIM-SSM	Protocol Independent Multicast Source Specific Multicast
PPP	Point to Point Protocol
PVST	Per Vlan Spanning Tree
QoS	Quality of Service
RO	Read Only
RW	Read Write
SAA	Service Assurance Agent
SDEE	Security Device Event
SFP	Small Form-Factor Pluggable
SNMP	Simple Network Management Protocol
SOC	Security Operation Center
SPOC	Single Point of Contact
SSH	Secure Shell
SSID	Service Set Identifier
SSL	Secure Socket Layer
STP	Spanning Tree Protocol
SOW	Statement Of Work
SW	Software
TACACS	Terminal Access Controller Access Control System
TCP	Transmission Control Protocol
TK-Equipment	Turn Key Equipment
TrustSec	Cisco Technology for scalable access control
UDP	User Datagram Protocol

Term	Definition
UPS	Uninterrupted Power Supply
VCS	Video Communication Server
VGW	Virtual Gateway
VLAN	Virtual LAN
VN	Virtual Network
VPN	Virtual Private Network
VPNSC	VPN Solution Center
VRF	Virtual Routing and Forwarding
VSS	Virtual Switching System
VTP	Virtual Trunking Protocol
WAAS	Wide Area Application Services
WAE	Wide Area Application Engine
WAVE	Wide Area Virtualization Engine
WAN	Wide Area Network
WCCP	Web Cache Communication Protocol
WFQ	Weighted Fair Queuing
WLC	Wireless LAN Controller
ZFW	Zone Based Firewall

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